

Electrochemical Air Quality Sensors: Principles, Operation, and Applications

1. Introduction

Modern air quality monitoring systems, in addition to well-established analytical methods, also rely on sensors capable of detecting gases and particles at very low concentrations. These sensors are used in environmental stations, industrial facilities, smart buildings, wearable devices, and smart-city networks.

The most well-known sensors used for air quality are:

Metal Oxide Semiconductor (MOS) Sensors :

- Change in electrical resistance of a metal oxide layer
- suitable for detection of: VOCs, CO, CH₄, H₂

NDIR (Non-Dispersive Infrared) Sensors

- Measurement of infrared light absorption
- CO₂, CH₄

Photoionization Detectors (PID)

- VOCs, BTEX compounds
- UV light ionizes gas molecules and generated ions are measured

Optical Particle Sensors (OPS)

- PM1, PM2.5, PM10
- Laser light scattering by airborne particles

Photoacoustic Sensors

- CO₂, CH₄, N₂O
- Absorbed light generates acoustic waves that are measured

Electrochemical Sensors

- CO, NO₂, SO₂, O₃, H₂S, NH₃
- Redox reactions generate an electrical current proportional to gas concentration

Photoacoustic Sensors

- CO₂, CH₄, N₂O
- Absorbed light generates acoustic waves that are measured

Catalytic Bead Sensors (Pellistors)

- Combustible gases (CH_4 , propane, butane)
- Heat generated by catalytic combustion changes resistance

Thermal Conductivity Sensors (TCD)

- H_2 , CH_4 , binary gas mixtures
- Measurement of changes in thermal conductivity

Flame Ionization Detectors (FID)

- Hydrocarbons, VOCs
- Ionization of compounds in a hydrogen flame

Metal Electrolyte / Solid-State Sensors

- O_2 , NO_x , automotive emissions
- Ionic conduction through a solid electrolyte

2. Electrochemical Air- Quality Sensors

Among the various sensing technologies available today, electrochemical sensors are widely used to detect gaseous pollutants such as carbon monoxide, nitrogen dioxide, sulfur dioxide, ozone, and hydrogen sulfide. Their advantages include high sensitivity, low power consumption, compact size, and relatively low cost.

2.1. Fundamentals of Electrochemistry

Electrochemistry is the branch of chemistry that studies the relationship between chemical reactions and electrical energy. Electrochemical reactions are deeply ingrained in everyday life, powering batteries, protecting against corrosion, and even driving metabolic processes in the human body. In sensor applications, electrochemical principles are harnessed to detect and quantify specific chemical species such as gases, ions, or biomolecules.

Whenever electrons are transferred between chemical species, an electrical current may be produced or consumed. Electrochemical reactions typically occur at the interface between an electrode (a solid conductor) and an electrolyte (a liquid or solid that contains free ions and conducts electricity).

The foundation of electrochemistry lies in redox reactions.

Redox Reactions: Short for "reduction-oxidation" reactions, these involve the transfer of electrons from one substance to another.

- o Oxidation is the loss of electrons.
- o Reduction is the gain of electrons.

Because oxidation and reduction always occur together, they are collectively called oxidation-reduction reactions.

Electrochemical systems consist of electrodes and an electrolyte. The electrodes are conductors through which electricity enters or leaves an electrochemical system; they provide surfaces where reactions occur.

- Anode: Electrode where oxidation occurs.
- Cathode: Electrode where reduction occurs.

The electrolyte is a medium (often a liquid solution) containing ions that can move to carry electric current and complete the electrical circuit.

2.2. Electrochemical Cells and Their Relevance to Sensors

Electrochemical cells are devices in which electrochemical reactions take place. They are characterized by *cell potential* (voltage), defined as the difference in electric potential between two electrodes; it measures the tendency of electrons to flow from one electrode to the other.

Electrochemical cells are broadly divided into two categories:

1. Galvanic cell
2. Electrolytic cell

2.2.1. Galvanic Cell Operation

Galvanic cells are devices that convert chemical energy into electrical energy through a spontaneous redox reaction. The most common examples of galvanic cells are batteries.

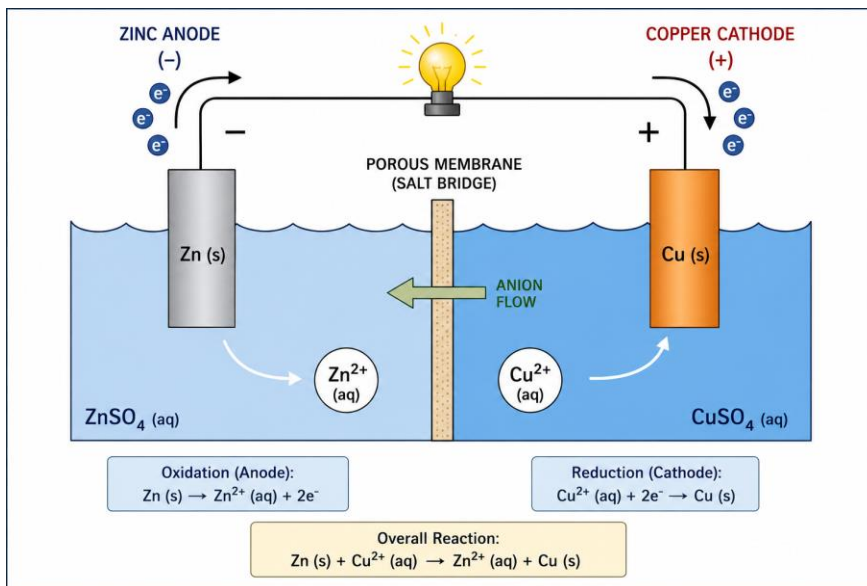
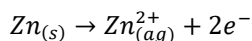


Figure 1. Galvanic cell, created by ChapGPT

The cell consists of two half-cells connected by an external circuit and separated by a porous membrane. One compartment contains a zinc electrode immersed in a zinc sulfate solution (ZnSO_4), while the other contains a copper electrode immersed in a copper sulfate solution (CuSO_4).

1. Oxidation at the Zinc Anode

The zinc electrode acts as the anode, where oxidation occurs. Zinc atoms lose electrons and enter the solution as zinc ions:



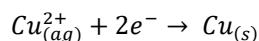
As a result, zinc metal gradually dissolves, the concentration of Zn^{2+} ions increases in the zinc sulfate solution, and the electrons are released into the external circuit.

2. Electron Flow Through the External Circuit

The electrons generated at the zinc anode cannot pass directly through the solution. Instead, they travel through the external wire, powering the light bulb shown in the circuit. The flow of electrons is: *Zinc anode* $\rightarrow$ *wire* $\rightarrow$ *light bulb* $\rightarrow$ *Copper cathode*. This movement of electrons constitutes the electric current produced by the cell.

3. Reduction at the Copper Cathode

The copper electrode acts as the cathode, where reduction takes place. Copper ions from the copper sulfate solution gain electrons arriving through the external circuit and are deposited as solid copper:



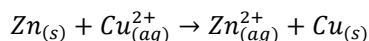
As a result, copper metal is deposited on the cathode surface, the concentration of Cu^{2+} ions in the solution decreases. The copper electrode gradually gains mass.

4. Ion Movement Through the Porous Membrane

The porous membrane separates the two solutions while allowing ions to move between them. As zinc oxidation produces positively charged Zn^{2+} ions, charge balance must be maintained. Therefore, negatively charged ions (anions) migrate through the membrane toward the zinc compartment. This ion movement prevents charge accumulation and allows the electrochemical reaction to continue.

5. Overall Cell Reaction

The combined oxidation and reduction reactions produce the overall cell reaction:



This spontaneous reaction generates electrical energy that can power external devices.

2.2.2. Electrolytic cell

Electrolytic cells consume electricity to drive reactions that would not occur spontaneously. The classic example of electrolytic cell is water electrolysis.

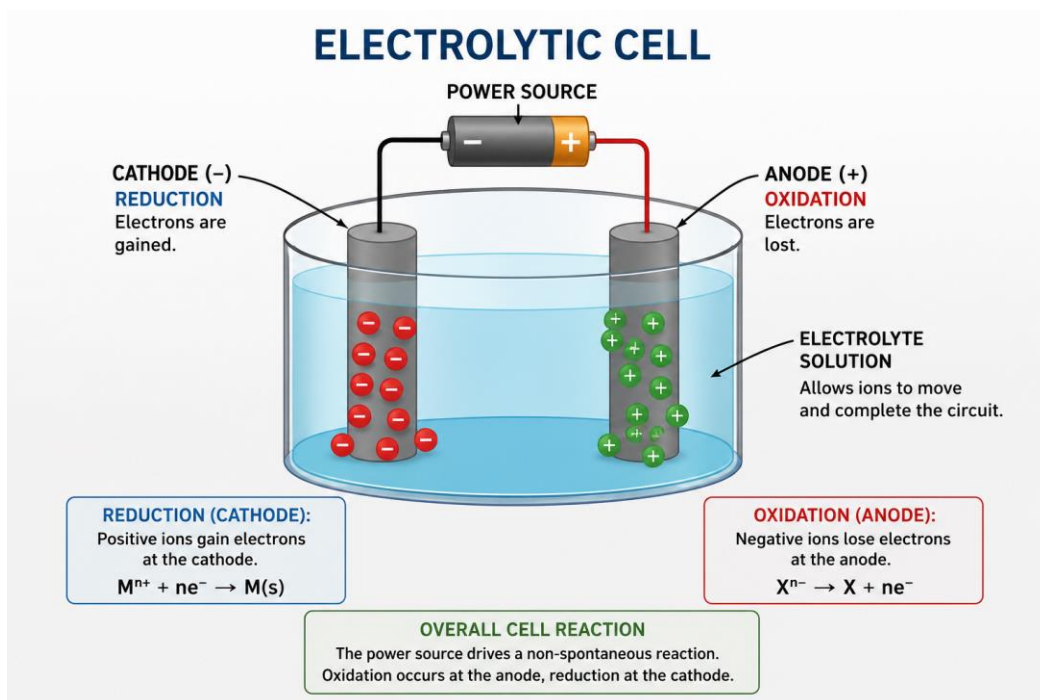


Figure 2. Electrolytic cell, created by ChatGPT

Unlike a galvanic cell, which generates electricity from a spontaneous reaction, an electrolytic cell consumes electrical energy supplied by a battery; it uses an external power source to drive a non-spontaneous chemical reaction.

The cell consists of two electrodes immersed in an electrolyte solution and connected to an external power source. The battery forces electrons to move through the circuit, enabling chemical reactions that would not occur naturally.

At the anode (positive electrode) oxidation takes place. Species present in the electrolyte lose electrons and become oxidized. The released electrons are drawn away by the external power source.

At the cathode (negative electrode), reduction occurs. Electrons supplied by the battery are delivered to the cathode, where ions in the electrolyte gain electrons and undergo reduction.

The electrolyte solution contains mobile ions that carry charge within the cell. Positively charged ions (cations) migrate toward the cathode, where they can gain electrons, while negatively charged ions (anions) move toward the anode, where they can lose electrons. This movement of ions completes the electrical circuit and maintains charge balance throughout the system.

The external power source is essential because it provides the energy required to overcome the natural thermodynamic barriers of the reaction. As a result, electrical energy is converted into chemical energy stored in the reaction products.

A common example of an electrolytic process is the electrolysis of water:

At the cathode (reduction): $2H_2O + 2e^- \rightarrow H_{2(g)} + 2OH^-$

At the anode (oxidation): $2H_2O \rightarrow O_{2(g)} + 4H^+ + 4e^-$

Overall reaction: $2H_2O \rightarrow 2H_{2(g)} + O_{2(g)}$

This principle is widely used in industrial electrolysis, metal electroplating, hydrogen production, material processing, and various electrochemical technologies.

Understanding these systems is essential because electrochemical sensors draw on many of the same principles as electrochemical cells. For sensor applications, both types can be relevant, but **galvanic** and **amperometric** systems are more commonly used. As mentioned, in a galvanic system, the current is the product of a spontaneous reaction, whereas in an amperometric configuration, the sensor maintains a constant electrode potential and measures the current generated by the electrochemical reaction.

2.3 Structure of an Electrochemical Gas Sensor

A typical electrochemical gas sensor contains several critical components shown in Figure 3. The working electrode is the location where the target gas reacts. The counter electrode completes the electrical circuit and balances the reaction. The reference electrode provides a stable potential to ensure accurate measurements. The electrolyte transports ions between electrodes. A gas diffusion barrier regulates the amount of gas entering the sensor and contributes to measurement linearity and stability. The interaction between these components determines the overall sensor performance.

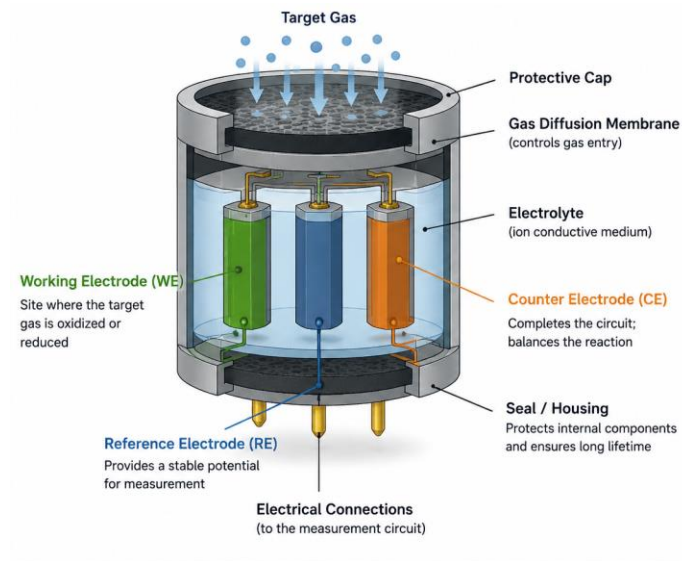


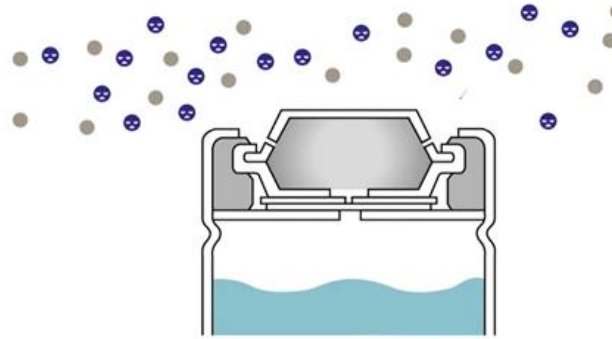
Figure 3. Electrochemical sensor compounds

2.4 Operating Principle of an Electrochemical Sensor

The operation of an electrochemical sensor can be divided into three stages.

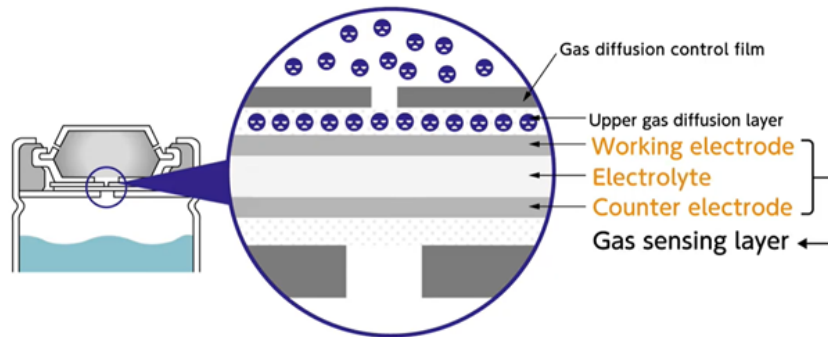
Step 1 – Gas Diffusion

When the target gas is present in the air, CO enters through the gas inlet.



Step 2 – Electrochemical Reaction

The target gas diffuses through a porous membrane and reaches the gas sensing layer, which consists of a working electrode, often containing a catalyst, a counter electrode, and an electrolyte that can transfer ions.



Step 3 – Signal Generation

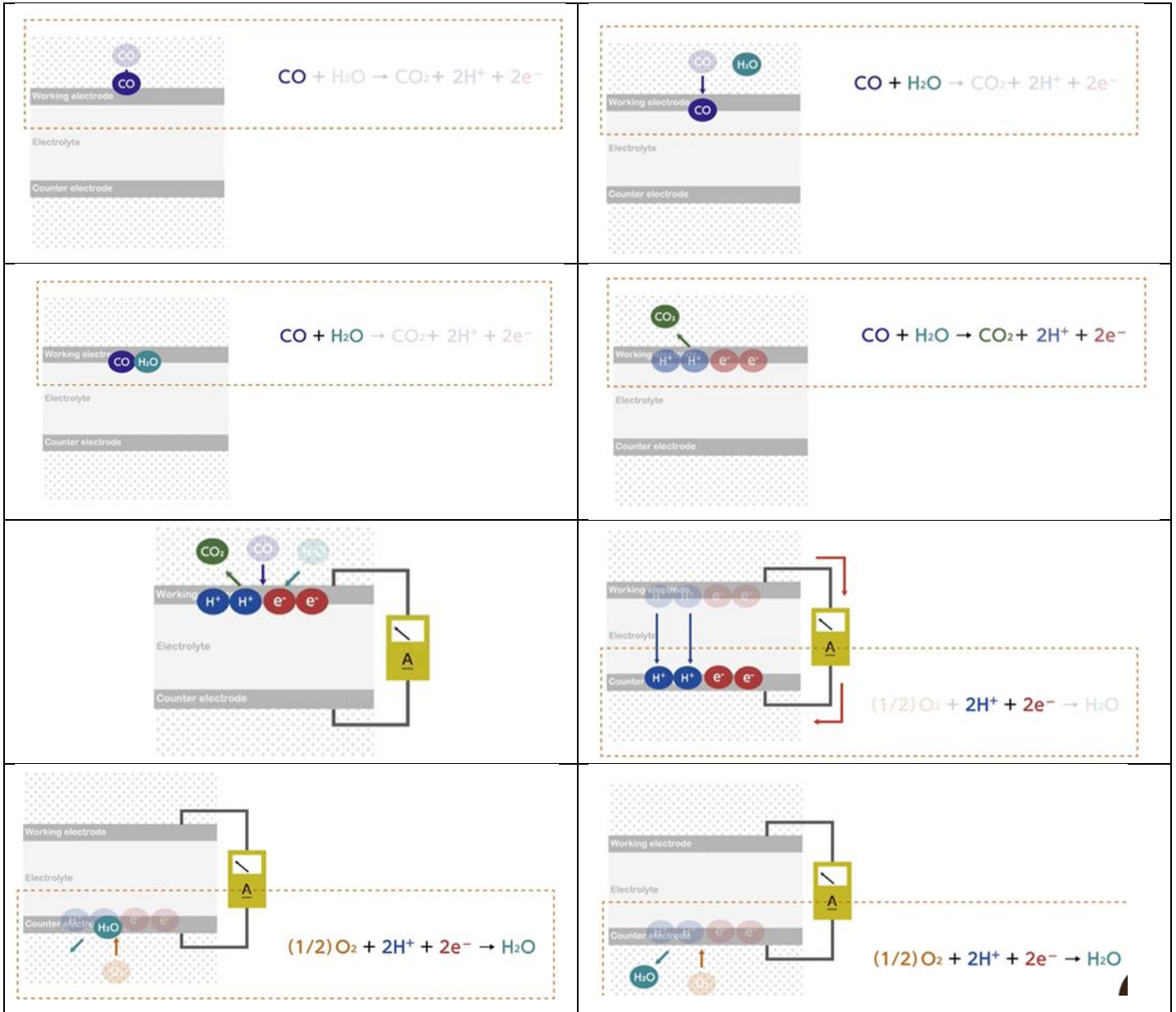
After the target gas reaches the sensor surface (working electrode), it undergoes oxidation or reduction, and the electrons generated by the reaction travel through the external circuit.

2.5 Examples of Electrochemical Gas Detection

2.5.1 Carbon Monoxide Detection

When CO reaches the working electrode (1), it reacts with water vapor (H_2O) in the air on the catalyst of the electrode (2) (3), generating carbon dioxide (CO_2) as well as hydrogen ions (H^+) and

electrons (e^-) (4). Hydrogen ions (H^+) generated on the working electrode flow through the electrolyte, and electrons (e^-) generated flow through the external wire, both to the counter electrode (5)(6). Hydrogen ions (H^+) and electrons (e^-) react with oxygen (O_2) in the air on the counter electrode, generating water (H_2O) (7), (8).



At the working electrode, carbon monoxide is oxidized to carbon dioxide. This reaction releases electrons that contribute to the sensor current. The magnitude of the current is proportional to the concentration of carbon monoxide in the surrounding air.

Working Electrode Reaction $CO + H_2O \rightarrow CO_2 + 2H^+ + 2e^-$

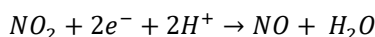
Counter Electrode Reaction $\frac{1}{2}O_2 + 2H^+ + 2e^- \rightarrow H_2O$

Overall Reaction $2CO + \frac{1}{2}O_2 \rightarrow CO_2$

2.5.2 Nitrogen Dioxide Detection

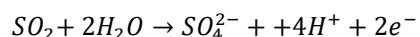
Nitrogen dioxide and sulfur dioxide are important atmospheric pollutants associated with combustion and industrial activities.

Nitrogen dioxide sensors typically operate through reduction reactions and can detect concentrations in the parts-per-billion range.



2.5.3 Sulfur Dioxide Detection

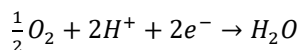
Sulfur dioxide sensors often rely on oxidation processes at the working electrode. These sensors are widely used in environmental monitoring stations and industrial emission monitoring systems.



In this reaction, sulfur dioxide (SO₂) is oxidized. Sulfate ions (SO₄²⁻) are produced. Protons (H⁺) are released into the electrolyte. Electrons (e⁻) are generated and flow through the external circuit. The generated current is proportional to the concentration of sulfur dioxide in the surrounding air.

Counter Electrode (Reduction)

To maintain charge balance, oxygen is typically reduced at the counter electrode:



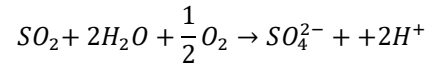
Overall Cell Reaction $SO_2 + 2H_2O + \frac{1}{2}O_2 \rightarrow SO_4^{2-} + 2H^+$

2.6 From Electrochemical Reaction to Gas Concentration

The electrical current generated by an electrochemical sensor is measured and converted into a gas concentration value. For many electrochemical gas sensors, the measured current is directly proportional to the concentration of the target gas, enabling accurate quantitative analysis.

The theoretical relationship between current and concentration is based on **Faraday's Law of Electrolysis**.

For a sulfur dioxide sensor, the oxidation reaction occurring at the working electrode is:



This reaction shows that each sulfur dioxide molecule releases two electrons during oxidation. According to Faraday's Law, the electrical current is related to the rate of electron generation:

$$I = zF \frac{dn}{dt}$$

where:

- I = current (A)
- z = number of electrons transferred per molecule (2 for SO₂)
- F = Faraday constant (96,485 C mol⁻¹)
- dn/dt = molar reaction rate (mol s⁻¹)

The molar reaction rate is proportional to the gas concentration:

$$\frac{dn}{dt} = KC$$

where:

- K = proportionality constant
- C = gas concentration

Substituting this expression into Faraday's Law gives:

$$I = zFKC$$

Rearranging:

$$C = \frac{I}{zFK}$$

Since the terms z, F, and K remain constant for a given sensor design, they are combined into a single parameter known as the sensor sensitivity:

$$S = zFK$$

Therefore:

$$C = \frac{I}{S}$$

where:

- C = gas concentration
- I = measured current
- S = sensor sensitivity (typically expressed in μA/ppm or nA/ppb)

In practice, the sensitivity constant (S) cannot be calculated accurately from theory alone because it depends on factors such as:

- Electrode surface area
- Catalyst properties
- Electrolyte composition
- Gas diffusion rate
- Membrane characteristics
- Sensor geometry

Therefore, the sensitivity is determined experimentally during sensor calibration. The sensor is exposed to known gas concentrations, and the resulting current is measured. The slope of the calibration curve corresponds to the sensor sensitivity. Once S is known, the sensor electronics can calculate gas concentration directly from the measured current using:

$$C = \frac{I}{S}$$

This equation is the one most commonly implemented in commercial electrochemical gas sensors.

2.7 Environmental and Operational Factors Affecting Sensor Performance

The performance of electrochemical sensors is influenced by several environmental and operational factors. Understanding these factors is essential for ensuring accurate measurements and reliable long-term operation.

Temperature

Temperature affects multiple aspects of sensor performance, including reaction kinetics, gas diffusion rates, and electrolyte conductivity. Higher temperatures generally accelerate electrochemical reactions and increase gas diffusion, while lower temperatures can reduce sensor responsiveness. For this reason, most commercial sensors incorporate temperature compensation algorithms.

Humidity

Humidity influences the water content of the electrolyte and the mobility of ions within the sensor. Significant changes in relative humidity can alter the sensor baseline, affect sensitivity, and increase measurement uncertainty. Maintaining stable environmental conditions helps improve measurement accuracy.

Cross-Sensitivity

Although electrochemical sensors are designed to be selective, they may also respond to gases other than the target analyte. This phenomenon, known as cross-sensitivity, can lead to measurement errors. For example, an ozone sensor may partially respond to nitrogen dioxide, while a carbon monoxide sensor may show a response to hydrogen. Sensor manufacturers typically characterize these effects during calibration and provide correction factors when necessary.

Sensor Drift and Aging

Over time, sensor performance gradually changes due to electrolyte degradation, electrode aging, catalyst deactivation, and contamination. These processes can alter the sensor sensitivity and baseline signal, resulting in measurement drift. To maintain accuracy, periodic calibration is required throughout the sensor's operational lifetime.

Sensor Lifetime

Because of the gradual degradation of internal components, electrochemical sensors have a finite service life. Most commercial sensors typically operate reliably for approximately two to five years, depending on environmental conditions and exposure levels. Proper storage, routine calibration, and protection from extreme temperatures and humidity can help maximize sensor lifetime.

Advanced Compensation Methods

Modern air-quality monitoring systems increasingly employ advanced signal-processing techniques to improve sensor performance. Machine-learning algorithms can be used for drift correction, cross-sensitivity compensation, pattern recognition, and pollutant classification. These methods enhance measurement reliability and extend the practical usability of low-cost sensor networks.